

Day 85 Project

Machine Learning for Regime-Aware Position Sizing

1 Introduction

Traditional trading strategies often rely on fixed position sizes or simple signals such as long or short positions.

However, in real financial markets, optimal position sizing depends on market conditions, volatility, and regime dynamics.

This project introduces a machine learning framework for regime-aware position sizing, where exposure is dynamically adjusted based on predicted market regimes and risk.

2 Problem Formulation

Let returns be defined as:

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

We aim to model:

$$Position_t = f(X_t)$$

where X_t includes:

- past returns
- volatility
- momentum

3 Feature Construction

We define features:

$$X_t = (r_{t-1}, r_{t-2}, \sigma_t, m_t)$$

where:

- σ_t = rolling volatility
- m_t = momentum

4 Regime Detection

We define regimes using volatility:

$$Regime_t = \begin{cases} 0, & \text{low volatility (stable)} \\ 1, & \text{high volatility (stress)} \end{cases}$$

5 Machine Learning Model

We train a classifier:

$$\hat{Regime}_t = f(X_t)$$

This model predicts whether the market is in a high-risk or low-risk state.

6 Position Sizing Rule

We define position size as:

$$w_t = \begin{cases} 1, & \text{low volatility regime} \\ 0.3, & \text{high volatility regime} \end{cases}$$

This reduces exposure during uncertain periods.

7 Strategy Returns

$$R_t^{strategy} = w_t \cdot r_t$$

8 Conclusion

This framework integrates machine learning with risk-aware trading decisions.

Instead of focusing solely on prediction, it emphasizes capital allocation, which is central to real-world portfolio management.